

Hajar Salame

Data & AI Engineer — Industrial Systems Optimization & Data Science

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Skills

Programming: Python (Pandas, NumPy, SciPy, Scikit-learn, PySpark), SQL, Java

Artificial Intelligence & Data: Machine Learning (supervised / unsupervised), Deep Learning, time series analysis, NLP fundamentals, optimization, industrial AI applications, data pipelines & integration, data wrangling, Data Product Concepts.

Tools & Platforms: Palantir Foundry, Jupyter Notebook, GitLab, Git, Power BI, LaTeX, MATLAB

Languages: French (C1), English (C1 — IELTS 8), Arabic (native), Spanish (B1 — CEFR)

Soft Skills: Organization, rigor, teamwork, stakeholder communication, adaptability, project management, learning agility

Education

IMT Atlantique *Engineering Degree — Industrial Systems Optimization & Data Science* 2023 – 2026

- Operations Research & optimization: optimization problem modeling, constraint programming (Choco solver in Java), planning and scheduling.
- Data analysis & modeling: statistics, regression, hypothesis testing.
- Machine Learning fundamentals: supervised and unsupervised methods for data analysis and valorization.

Preparatory Classes for Engineering Schools, Lycée Méditerranéen, Martil (Morocco) 2021 – 2023

- Two years of intensive preparation for admission to top French engineering schools.

Professional Experience

Data Scientist / Data Engineer (End-of-Studies Internship) *CS Group, Toulouse — Airbus ZAOG Project* Apr. – Sept. 2026

- Collect, prepare, and structure heterogeneous sensor, industrial, and maintenance data (aircraft time series) using **Python, Spark, SQL, and Palantir Foundry**.
- Design and implement ML/DL **surrogate models** to replace physics-based rule models; benchmark new approaches against existing solutions and assess feasibility for cloud deployment.
- Build dashboards and interfaces for business teams; conduct ML/DL state-of-the-art benchmarking; contribute to predictive maintenance, eco-energy optimization, and intelligent automation projects.

Logistics & Data/Bi Engineer Intern *Tractafic Equipment France, Puteaux* May – Aug. 2025

- Developed Power BI dashboards from logistics and industrial operations data (flows, lead times, inventory) for visualization and decision support.
- Automated data collection and cleaning pipelines; contributed to modernizing the Data/Bi architecture and cross-functional data integration.

Academic Projects

Learning Path Analysis *Project with Lab-STICC (ongoing)* Sept. 2025 – Mar. 2026

- Applied predictive methods and graph analysis to model and analyze temporal behaviors of apprentices.
- Co-authoring a scientific publication in collaboration with research faculty.

SMS & Email Spam Classification — NLP Pipeline *IMT Atlantique* Nov. – Dec. 2025

- Built an end-to-end NLP pipeline for spam detection across SMS and email: text preprocessing, feature engineering (bag-of-words, tokenization), model training, and evaluation (*Python, scikit-learn, NLTK*).
- Ran domain-specific and cross-domain experiments (SMS-only, email-only, combined, transfer learning); achieved ~96% accuracy on SMS and ~99% on email datasets with a modular, tested codebase.

Industrial Waste Optimization *Project with Reeverse Systems* Sept. – Dec. 2024

- Designed an optimization algorithm to reduce industrial material waste and standardize production; translated business requirements into a technical solution and presented results to non-technical stakeholders.

Engagement Associatif

- **World Atlantic Kelio Event organization:** managed logistics and coordinated all activities for an event hosting nearly 200 international students, demonstrating inclusion and cross-cultural project management.

Personal Interests & Hobbies

Sports: Badminton, rock climbing, and fitness. **Reading:** Science fiction and fiction. **Music:** Singing.